

FGR Method

Method Used

- The scans were downsampled using voxel downsampling with voxel length of 0.003. This parameter allows us to control the point density.
- FPFH extraction
- Fast Global Registration
- Point to Plane ICP refinement

Challenges faced

Randomness

- The FGR library used in open3d does some complex optimization as a result for the same code some runs produced perfect registration and some runs produced distorted image.

Incremental Merging

- This method uses incremental merging. That is bun05 is merged with bun000. now bun090 is merged with the combination of (bun000+bun045). As a result on merging bun180 to bun090 possibly due to the sharp 90 degree turn registration fails for bun180 and bun270.

Observations

- Visually FGR output produced convincing results. ICP refinement made only tiny improvement in many mergings. Had I done the same without ICP I may have gotten a similar scan possibly.
- FGR library produced randomness for same code on different runs. When I checked found out that this was due to some optimization inside the library. Pair wise scans which I have sent before worked well for any pair in the dataset. Randomness creeps in only when merging the full dataset.
- During the merging process in some cases the Fitness score produced by FGR were very low (0.20, 0.39, 0.32). But ICP boosted the fitness to above 0.6 range benefitting the registration. I am tabulating this below.

Scan	FGR	ICP
bun045	0.92	0.91
bun090	0.20	0.65
bun315	0.86	0.84
top2	0.39	0.69
top3	0.85	1.00
chin	0.76	0.76

Scan	FGR	ICP
ear_back	0.32	0.99

Problem left

The scans bun180 and bun270 could not be merged with the scan without causing distortion. as a result some areas which were to be filled by these were left with holes. Need to figure out a solution for this.I am attaching the images below.



